Application requirements example: Car Rental System

1. Add a description of the application you are implementing.

   A web-based Car Rental System that lets customers browse, reserve, and rent vehicles; and enables staff to manage fleet, customers, rentals, payments, and reports. It exposes a REST API secured with token-based authentication. The system generates rental agreements and invoices as PDFs/printables.

   Programming language used: Java
   Frameworks/databases: MySQL, Spring Boot, …

2. Key functionalities of the application
   a. A register/login system (username/password, token-based authentication)
   b. Ability to create read update and delete some objects (cars, customers, employees) in the application (via HTTP REST API).
   c. Ability to generate reports (car rental agreement, invoices)
   d. Filter by branch, date range, category, price, transmission, fuel type.
   e. Filter by daily rate + optional fees (GPS, child seat, one-way fee), deposits, insurance packages.
   f. …

3. Description of entities and relationships
   a. Class Vehicle
      i. Id
      ii. Registration_no
      iii. Make
      iv. Model
      v. fuel_type
      vi. Transmission
      vii. …
   b. Class Person
      i. Id
      ii. Ssn
      iii. Role = (1 for employees, 0 for customers, …)
      iv. …
   c. Class Rental
      i. Customer_id
      ii. Vehicle_id
      iii. Start_date

iv. end_date

Create an UML Class diagram
Any other kind of UML diagrams or other diagrams